

Workforce Management Platform

A plain-language guide to what the website does, page by page

1. What This Platform Is

This is an internal tool for a company to manage its workforce day to day. It answers the practical questions a business actually asks:

- Who is working today, and who is absent?
- Who is scheduled to work, versus who actually showed up?
- How many hours did each employee work, and how much of that was overtime?
- Who is on approved leave, and who has a leave request waiting for approval?
- How much is labor costing the company, per department?
- What privileged changes has anyone made recently (as a security record)?

It is built as a single, self-contained web application (Python/Flask on the backend, PostgreSQL as the database, plain HTML/CSS/JavaScript on the frontend) with no dependence on outside services. Everything you see in the browser is generated by the server, checked against real database records — nothing on any page is fake or placeholder data.

2. The Three User Roles

Every person who logs in has exactly one role, and the website looks and behaves differently for each:

Role	What they can do
Admin	Full access to everything: all employees, all departments, pay rates, labor cost figures, the audit log, and every management action. There is normally one Admin per organization — typically an owner or HR/operations lead.
Manager	A supervisor responsible for one or more specific departments. They can manage schedules, attendance, and leave for their own people, and see department-level cost totals — but never an individual employee's pay rate or exact cost, and never the audit log. This boundary is enforced by the server itself, not just hidden in the interface.
Employee	Self-service only. An employee can see their own schedule, clock in and out, view their own worked hours, and submit their own leave requests. They cannot see or affect anyone else's data.

This role boundary is not just a matter of what buttons are shown — every single action is re-checked on the server before it happens. Even if someone tried to bypass the interface entirely (e.g. by typing a web address directly), the server refuses anything their role doesn't allow.

3. The Core Idea: Planned Work vs. Actual Work

The single most important distinction in this system, and the one most workforce tools get wrong, is this:

- **Schedule** represents what was *planned*. A shift says "this employee is expected to work from 9:00 to 17:00."
- **Attendance** represents what *actually happened*. A clock-in/clock-out record says "this employee was physically working from 8:57 to 17:21."

These are always kept as two separate, distinct kinds of records. The system never assumes a schedule was followed exactly, and never treats a clock-in as automatically matching a shift. This is what allows the platform to answer real questions like "was this employee late," "did this shift go unstaffed," or "how much overtime did this cost," instead of just showing one blurred-together number.

4. How the Pieces Connect

Conceptually, information flows through the system like this:

Schedule (planned) + Attendance (actual) → Working Hours → Overtime → Labor Cost

- **Working Hours** compares what was scheduled against what was actually worked, for a given employee and day.
- **Overtime** is calculated automatically from the actual worked hours, using tiered rules (extra pay multipliers once daily and weekly thresholds are crossed) — the same rule for every hourly employee, applied consistently, never guessed at.
- **Labor Cost** combines worked hours, overtime multipliers, and an employee's hourly pay rate into a real monetary figure, calculated precisely (never using imprecise decimal math, which matters when money is involved).

Leave requests interact with this picture too: an approved leave request is checked against the schedule so the system never contradicts itself by showing someone as both "scheduled to work" and "on approved leave" at the same time.

5. A Sensitive Boundary: Who Can See Pay and Cost

Because pay rates and individual labor cost are sensitive financial information, the platform enforces one firm rule throughout: **a Manager can see department-level cost totals, and can see how many overtime hours an individual worked, but can never see an individual employee's hourly pay rate or their individual dollar cost.** Only an Admin can see that level of detail, and only on one specific, clearly separated page. This rule is enforced by the server on every relevant page, not just hidden from view.

6. Page-by-Page Guide

Login

Who can see it: Anyone (not yet signed in).

The entry point to the site. A short number of repeated wrong password attempts temporarily locks the account, to protect against someone trying to guess a password.

Dashboard

Who can see it: Everyone, but the content is completely different depending on role.

Admin / Manager view — an operational overview of "what's happening right now":

- Headline numbers: total employees, how many are working today, how many are absent, how many are on leave.

- A "Requiring Attention" panel that surfaces real operational problems needing a human's action

- attendance records flagged for review, or employees missing pay-rate/policy setup.
- Detail tables: exactly who is working, who is absent, who is on leave, and department-level coverage and labor cost.

Employee view — a personal dashboard:

- A prominent "Clock In" / "Clock Out" control, front and center, since this is the single most-used action for an employee.
- Upcoming shifts, recent worked hours (compared against what was scheduled), and pending leave requests.

Employees

Who can see it: Admin and Manager.

The full employee directory: name, department, and employment status (active, inactive, or terminated). Admins can add new employees. Clicking an employee opens their detail page.

Employee Detail

Who can see it: Admin (any employee), Manager (only employees in departments they manage).

A single employee's profile: contact details, department, employment status, and hire date. Admins and managers can edit these details here. Admins additionally see direct links to that employee's pay rate history and their individual labor cost breakdown, and can terminate an employee's record.

Pay Rate

Who can see it: Admin only.

An individual employee's hourly pay rate history over time, and a form to set a new rate effective from a given date. This is treated as the most sensitive page in the system — a Manager cannot reach it under any circumstance.

Departments

Who can see it: Admin and Manager.

The list of departments in the organization, with active/inactive status. Admins can create new departments, edit existing ones, or deactivate one that's no longer in use (deactivating never deletes historical records, it just stops it being usable for new assignments).

Schedule

Who can see it: Everyone. Employees see only their own shifts; Admin/Manager see everyone's (Manager limited to their departments).

The planned work calendar. A shift has a status: *draft* (not yet finalized), *published* (confirmed and visible to the employee), or *cancelled*. Admins and Managers can create shifts, assign them to an employee, publish them, or cancel them.

Attendance

Who can see it: Everyone. Employees see only their own records; Admin/Manager see everyone's (Manager limited to their departments).

The actual record of who worked when. Anyone can clock themselves in and out from here (and it's also available on the Employee Dashboard for convenience). Admins and Managers can correct a record if something was entered wrong, always with a required written reason so the change is traceable. An attendance record that was left open too long without a clock-out is automatically flagged for review, and can only be resolved by an Admin or Manager's correction — never by simply clocking out, since that would defeat the purpose of the flag.

Leave

Who can see it: Everyone. Employees see only their own requests; Admin/Manager see everyone's (Manager limited to their departments) and can approve or reject.

Every employee can submit a leave request here and track its status: *pending*, *approved*, *rejected*, or *cancelled*. If a pending request overlaps with an already-published shift, the system flags that conflict explicitly for the approving manager to see, rather than silently ignoring it.

Overtime Report

Who can see it: Admin and Manager.

For a chosen department and date range, shows each employee's total overtime hours. Deliberately shows hours only — never a pay rate or a cost figure, consistent with the sensitive-information boundary described above.

Hours Trend

Who can see it: Admin and Manager.

For a chosen department and date range, shows total worked hours per day as a simple bar comparison, to answer "are working hours going up or down over time" at a glance.

Labor Cost

Who can see it: Admin and Manager.

For a chosen department and date range, shows the total labor cost — a single aggregate figure, never broken down by individual employee on this page. If any employees are missing pay-rate or policy setup, that gap is called out explicitly rather than silently producing a wrong total.

Labor Cost — Employee Breakdown

Who can see it: Admin only.

The one page in the entire system that shows an individual employee's hourly rate and their exact cost, broken down day by day and category (regular time, daily overtime, weekly overtime). Never reachable by a Manager.

Audit Log

Who can see it: Admin only.

A complete, permanent record of privileged actions across the whole organization — logins, pay rate changes, leave decisions, employee changes, and more — each entry showing who did it, when, and from what network address. Records are never edited or deleted once written. Passwords and pay-rate values themselves are never stored in this log, only the fact that an action happened.

Recent Activity

Who can see it: Admin only.

A lighter, easier-to-read version of the audit log — the most recent events only, for a quick glance rather than a full investigation. Reachable directly from the sidebar on every page, and links through to the full Audit Log for the complete history.

7. How the System Protects Data

- Passwords are never stored in readable form — only a one-way cryptographic hash that cannot be reversed.
- Every page and every action re-checks the signed-in user's role and organizational scope on the server, every single time — the interface never being shown a button is not what keeps data safe, the server's own check is.
- A Manager can never see another department's data, even by directly typing a web address for it.
- All money calculations use exact decimal arithmetic, never the kind of approximate math that can lose a cent here and there.
- Every database rule that matters (an employee can't be clocked in twice at once, a shift can't overlap another for the same employee, a leave request can't be approved and rejected at the same time) is enforced by the database itself, as a last line of defense, not only by the application code.

8. Technical Foundation (For Reference)

Backend: Python, Flask, SQLAlchemy, PostgreSQL. Frontend: server-rendered HTML templates, plain CSS, and a small amount of vanilla JavaScript — deliberately no external frontend framework, keeping the system simple to maintain and easy to reason about. The whole application is organized as a single, well-structured codebase (a "modular monolith") rather than a scattered collection of separate services, which keeps it easier to understand, test, and secure as it grows.